Linear Regression Analysis of Climate Data to Identify Drivers of Climate Change

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Project Description

This project is a regression-based analysis of data points gathered from multiple sources, intended to investigate the most influential causes of climate change. After collecting anthropogenic and climate data from reputable scientific and government sources, the project includes multiple data preprocessing techniques, model selection and development, feature ranking, and data visualization. Finally, this report contains discussion about the various results of the project, as well as explanations of the aforementioned intermediate steps.

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Step 1: Data Collection

I procured multiple datasets for this project, with most coming from the required sources quoted in the assignment brief, though I also used a dataset that I found online. References to all used datasets can be found in the References section. After reading the data from all the datasets, I merged them into one large dataset using methods from the Pandas library in Python, using the shared *'year'* column present in all the datasets as a merge anchor.

List of used datasets

1. NASA GISS Global Surface Temperature Anomaly Measurements ("GLB.Ts+dSST.csv")
2. NOAA GML Atmospheric Methane Measurements ("ch4_annmean_gl.csv")
3. NOAA GML Atmospheric CO_2 Measurements ("co2_annmean_mlo.csv")
4. NOAA GML Atmospheric N_2O Measurements ("n2o_annmean_gl.csv")
5. Our World in Data CO_2 Emissions Database ("owid-co2-data.csv")
6. NOAA GML Atmospheric SF_6 Measurements ("sf6_annmean_gl.csv")
7. The Smithsonian Global Volcanism Program Yearly Eruptions Database ("volcanoes.csv")

Step 2: Data Preprocessing

From the massive, merged dataset, I selected the most important and interesting features (columns). The most important of these was the 'J-D' (annual temperature anomaly) column from the NASA GISS Global Surface Temperature Anomaly Measurements dataset as it functioned as the target/dependent variable for this analysis.

Because the original datasets covered wildly different stretches of time, many cells contained 'NaN' values. I decided to drop rows that had a large number of missing values, using a threshold to drop rows with at least 3 'NaN' values. At this point, I also renamed the columns in the current dataset to make programming easier and the data more readable.

As there were still some rows that had missing values, I decided to use imputation to fill them with values. I used the Pandas linear interpolation method. Interestingly, because most of the missing values occurred at the head of the dataset, the interpolation was functionally equivalent to performing a backfill on the affected columns.

As requested in the brief, I engineered some new features from the data already present: 'CH_4 emissions growth percentage' from 'methane emissions', 'atmospheric CO_2 growth percentage' from 'atmospheric CO_2 mean', and 'atmospheric CH_4 growth percentage' from 'atmospheric CH_4 mean'.

Step 3: Model Development

To perform analysis on the preprocessed dataset, I trained a linear regression model to target the 'Temperature Anomaly' variable (formerly 'J-D') using all other selected and engineered features. I performed a minimal amount of analysis of model performance, satisfied with just its accuracy score. The model had an R^2 score of 0.866, showing that the features included explained an impressive 86.6% of the variance in the data.

Step 4: Root Cause Identification

One of the most important parts of the project was determining which of the features in the dataset were the most correlated with the dependent value ('Temperature Anomaly'). Using the Select k-Best feature ranking technique, I was able to ascertain the features that explained the greatest amount of variance in the dependent variable. These were, in alphabetical order: 'atmospheric CO_2 mean', 'atmospheric CH_4 mean', 'atmospheric SF_6 mean', 'CO_2 emissions', 'methane emissions', 'nitrous oxide emissions', and 'primary energy consumption'.

To further investigate the influence of anthropogenic ('human-driven') and natural factors, I tried two separate regression models solely on anthropogenic and natural factors respectively. The model trained solely on anthropogenic factors (greenhouse gas emissions and energy usage) had an R^2 score of 0.919, while the model trained solely on natural factors (volcanic activity, especially) had an R^2 score of 0.783.

Interpretation of Root Cause Analysis

From the analysis above, the obvious conclusion is that greenhouse gas emissions (especially CO_2 and CH_4) are the greatest drivers of climate change. Aerosol levels (SF_6) is also a noticeable driver of climate change, though to a much lesser extent (see Figure 2). Comparing human-driven and natural factors clearly shows, at least from the dataset used in this project, that human-driven factors are much

more closely correlated with global temperature anomalies when compared to natural factors. That said, the correlation between natural factors of climate change was not as low as I was expecting it to be, though it could be argued that the main natural factor included in this analysis (volcanic activity) may also be affected by human activity.

Step 5: Data Visualization

All figures and plots in this section were created directly from the dataset collated in this project and created using the Matplotlib and Seaborn libraries in Python.

Figure 1: Greenhouse Gases Time Trends vs. Temperature

Figure 1a

Atmospheric CO_2 Mean vs. Temperature Anomaly Over Time

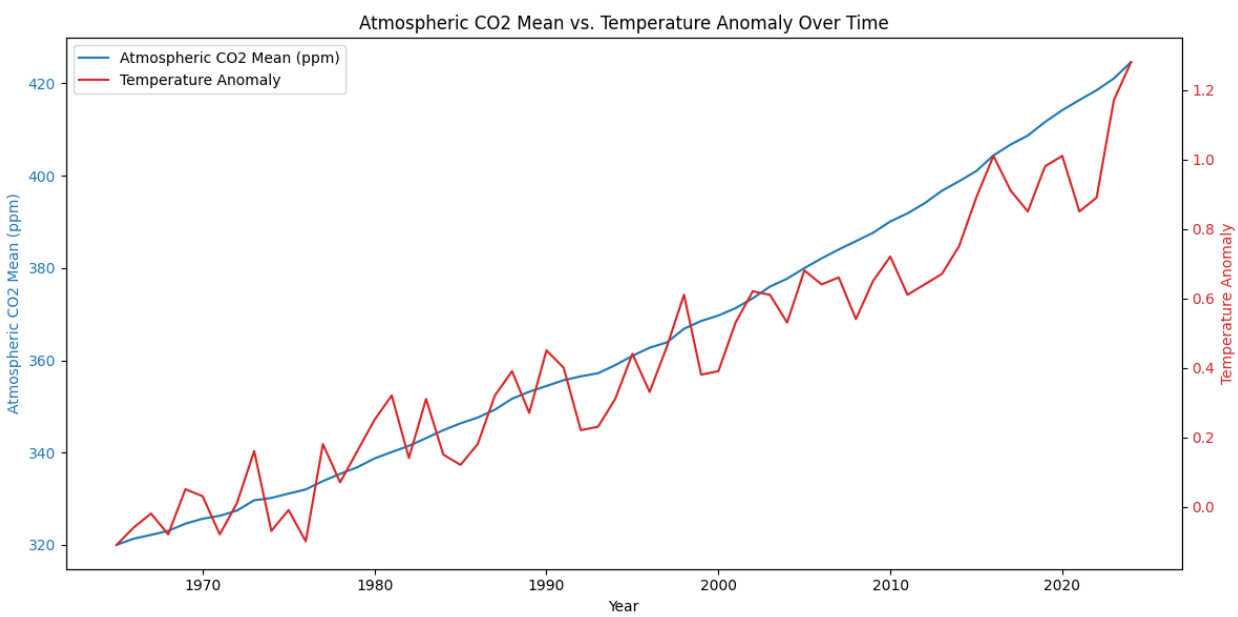

Figure 1b

Atmospheric CH_4 Mean vs. Temperature Anomaly Over Time

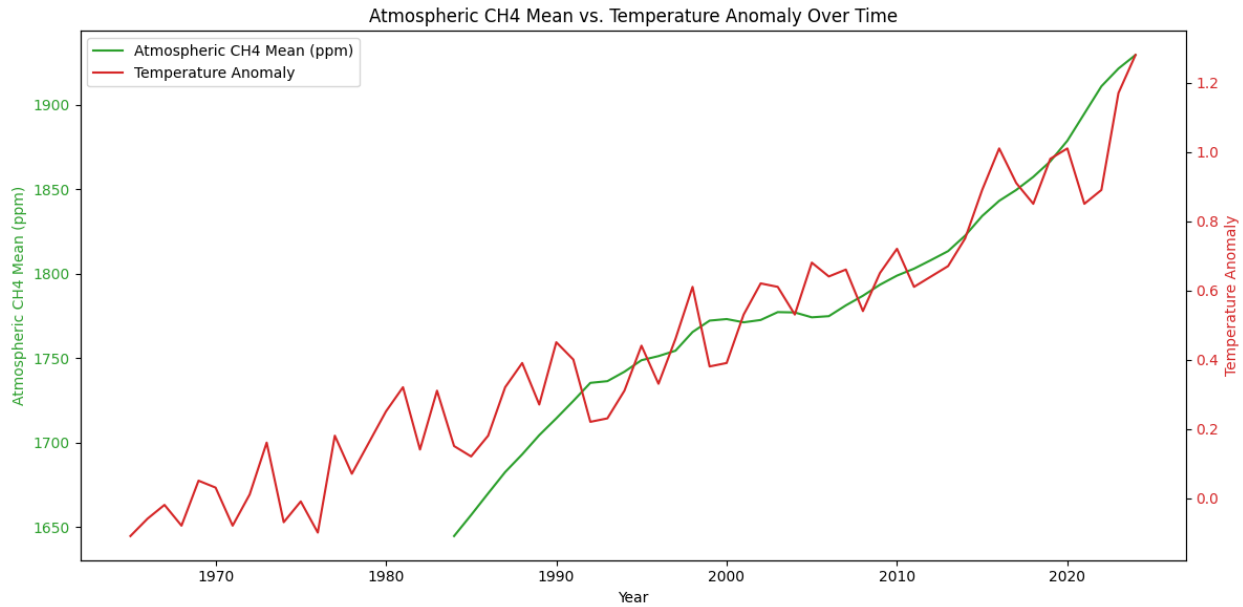

Figure 1c

Atmospheric N_2O Mean vs. Temperature Anomaly Over Time

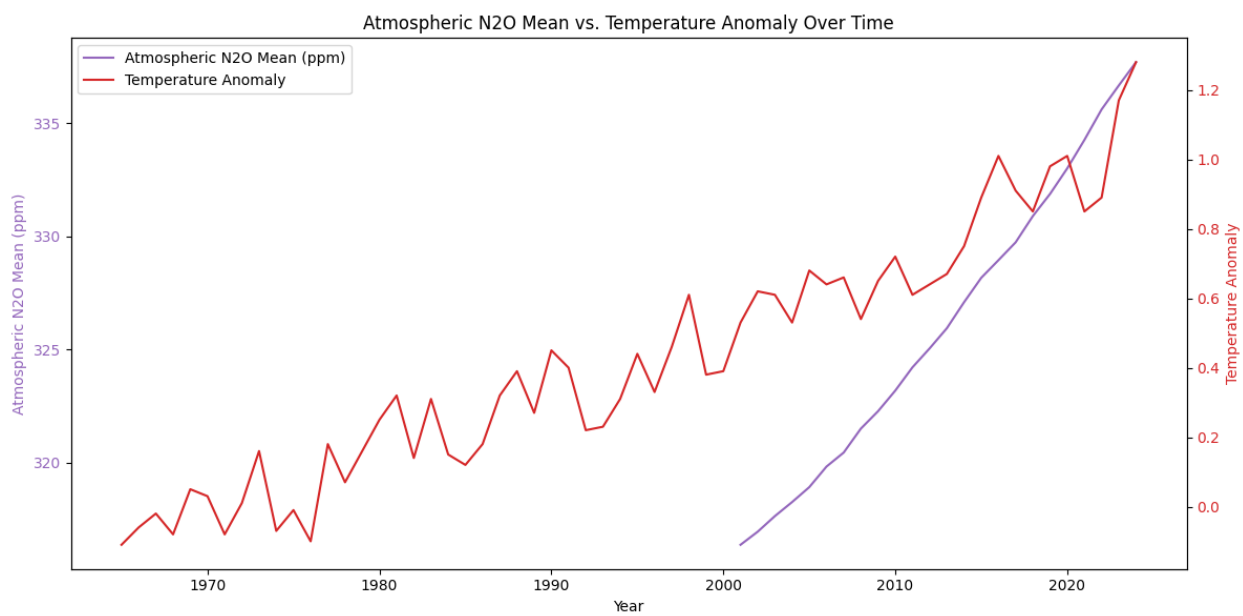

Figure 1d

Atmospheric SF_6 Mean vs. Temperature Anomaly Over Time

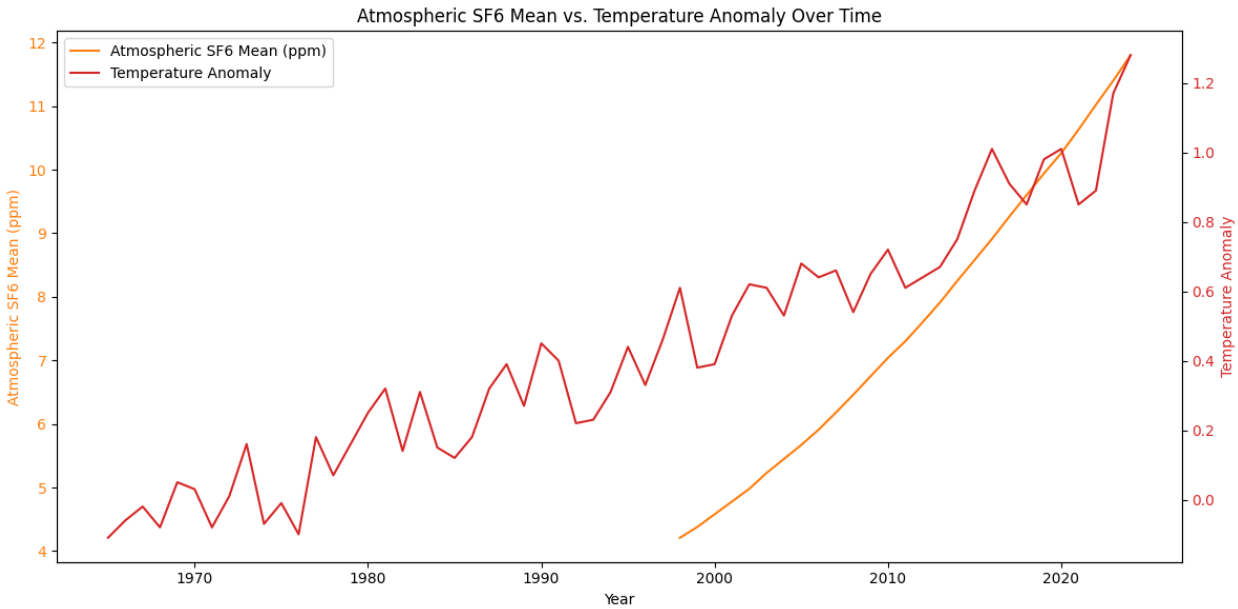

Figure 2: Feature Importance Bar Chart

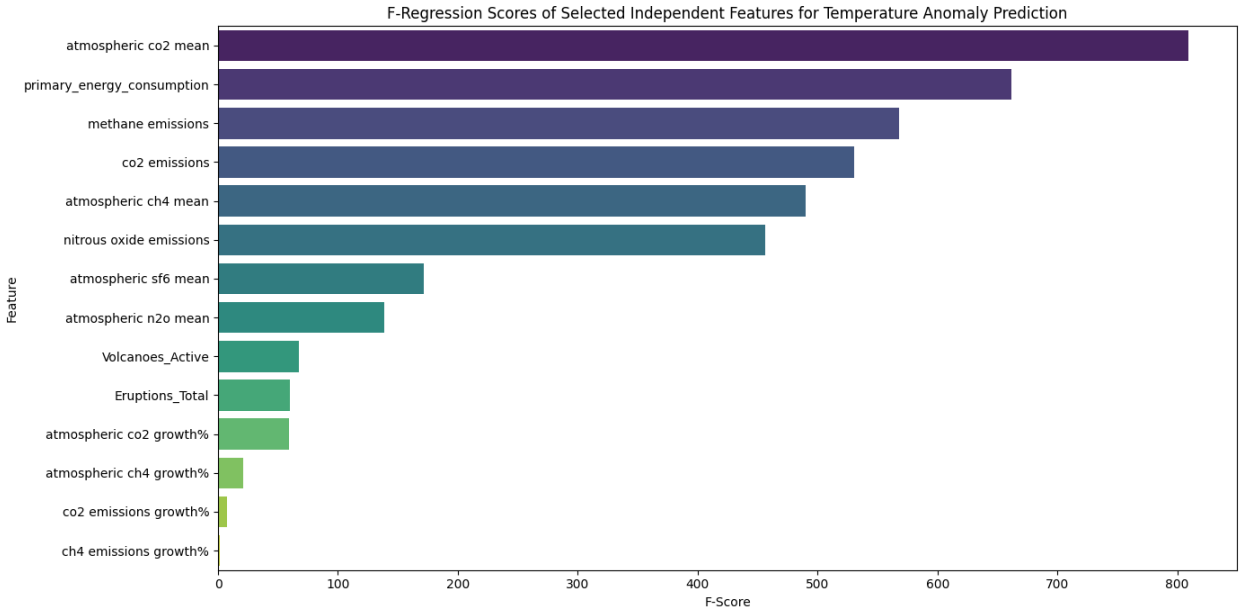

Figure 3: Top features scatter plots with regression line

Figure 3a

Atmospheric CO2 Mean vs. Temperature Anomaly

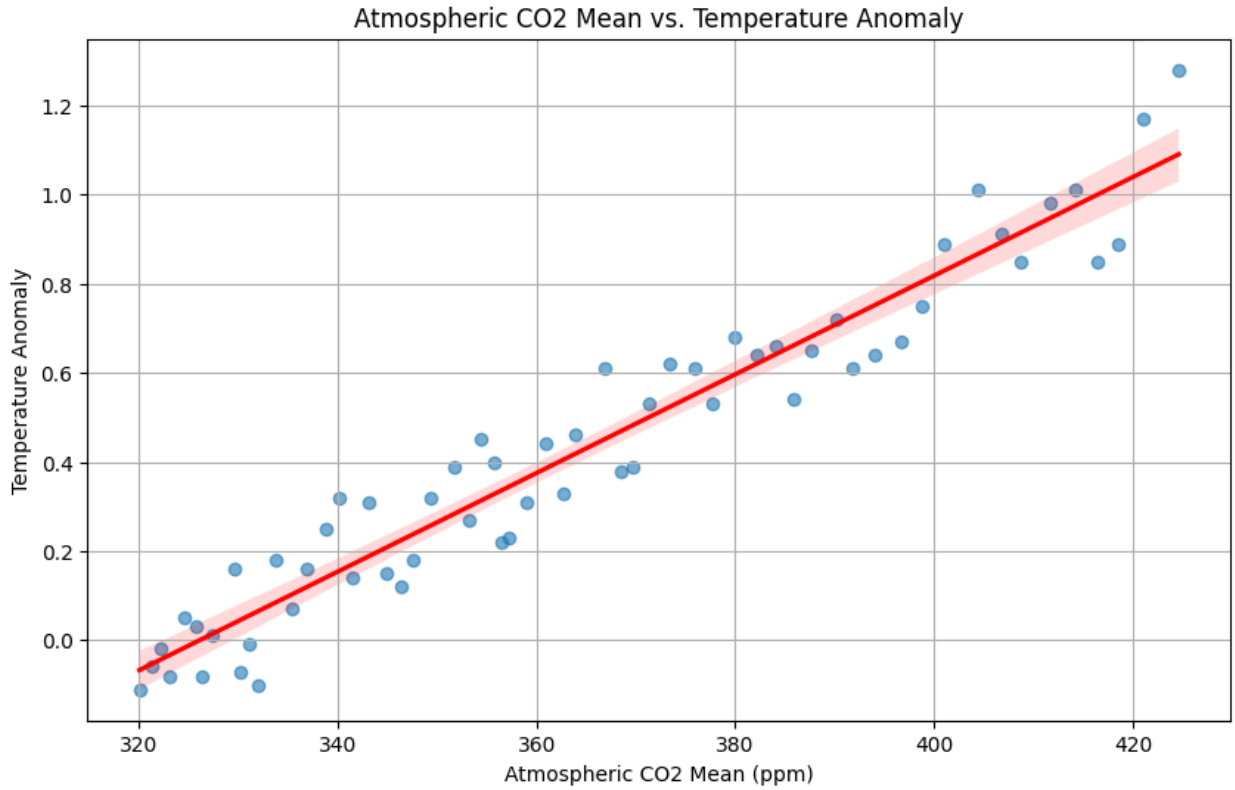

Figure 3b

Primary energy consumption vs. Temperature Anomaly

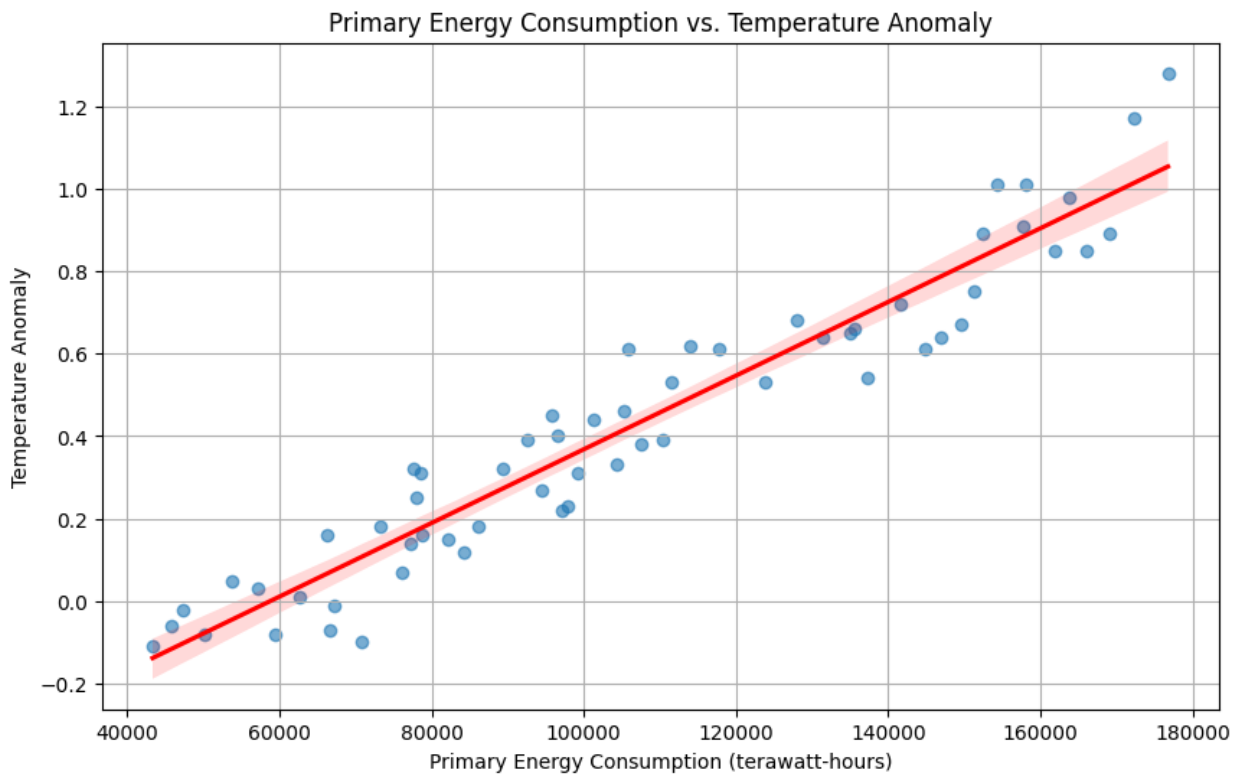

Figure 3c

Methane emissions vs. Temperature Anomaly

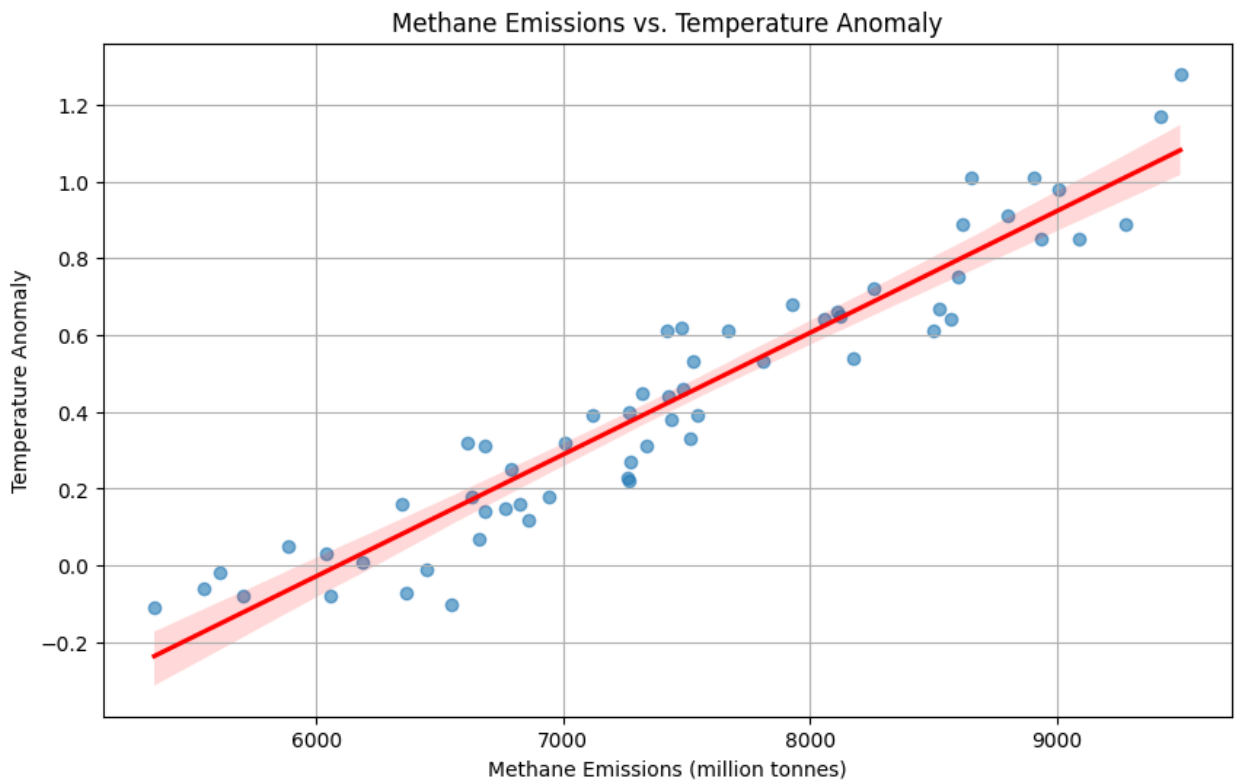

Figure 3d

CO_2 emissions vs Temperature Anomaly

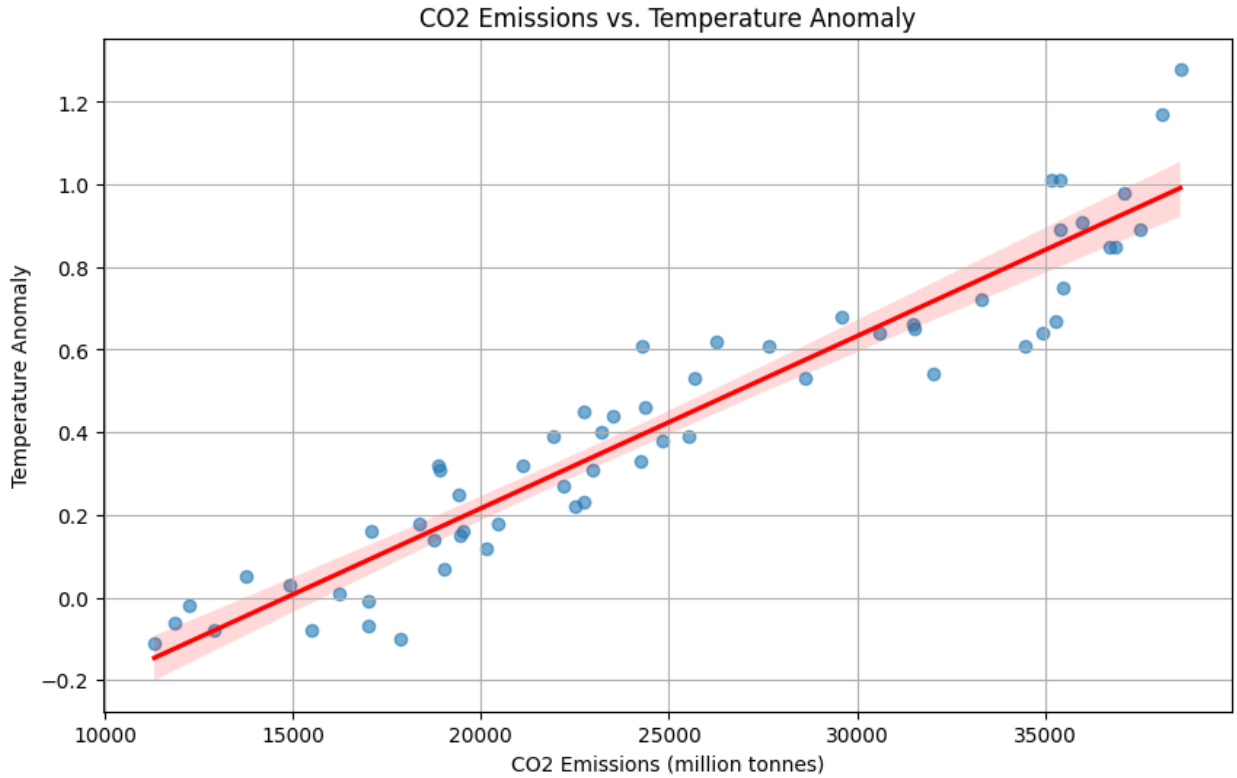

Figure 3e

Nitrous oxide emissions vs. Temperature Anomaly

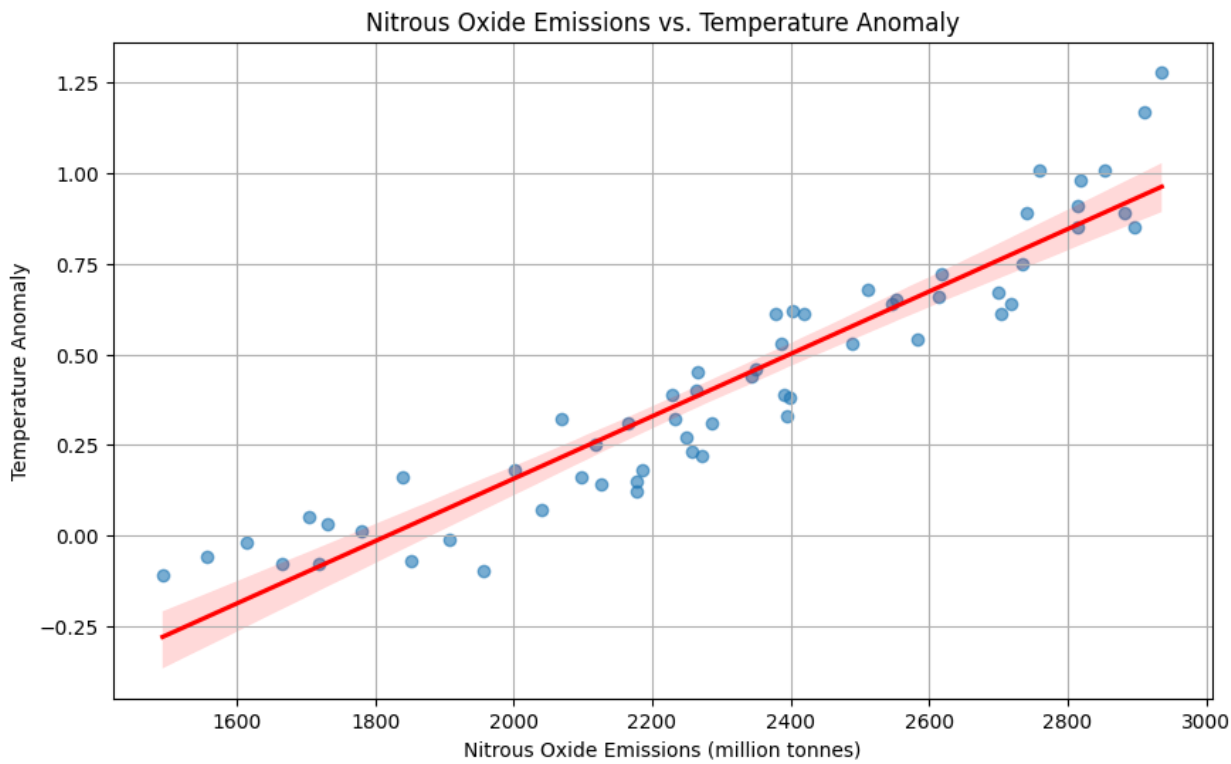

Table 1: Example rows of merged data after preprocessing

	year	atmospheric co2 mean	atmospheric ch4 mean	atmospheric n2o mean	atmospheric sf6 mean	co2 emissions	co2 emissions growth%	methane emissions	nitrous oxide emissions	primary_energy_consumption	Eruptions_Total
270	2020	414.21	1878.72	333.01	10.26	35158.230469	-5.199564	8903.342773	2852.966797	157993.890625	72.0
271	2021	416.41	1894.81	334.27	10.63	36866.863281	4.859841	9089.302734	2895.432373	166043.500000	82.0
272	2022	418.53	1910.94	335.62	11.02	37527.773438	1.792693	9278.555664	2881.229492	169061.531250	85.0
273	2023	421.08	1921.43	336.68	11.40	38094.039062	1.508927	9416.929688	2911.154053	172238.781250	75.0
274	2024	424.61	1929.47	337.71	11.80	38598.578125	1.324451	9498.914062	2935.778564	176737.093750	73.0

Discussion and Conclusion

The results of this project show that anthropogenic factors are the greatest drivers of climate change. My analysis particularly singled out the amount of CO_2 in the atmosphere, global energy consumption, and methane and CO_2 emissions as the most significant contributors. This intuitively makes sense: the more greenhouse gases we emit, the more the planet heats up. The high contribution of energy consumption is also understandable: fossil fuels are still the primary source of global energy, so energy consumption correlates highly with emissions. In these ways, the results of the analysis performed during this project completely agree with established climate science.

The initial data collection process was interesting, involving learning how to skip rows that contain comments and how to merge disparate datasets together using common columns. Data preprocessing was by far the most involved part of the process, requiring multiple techniques including dropping rows, data imputation, and column renaming for better readability. Model development was straightforward, using linear regression for the overall model as well as the human-driven and natural factors models, and k-Best Selection for feature engineering/root cause identification.

Using linear regression analysis for this sort of causal inference does have some limitations. Firstly, there are likely many variables that contribute to our dependent variable (temperature anomaly) that are not included in this analysis. Thus, there might be a bias that is overstating the importance of the independent variables that *are* present in the analysis. Some of these missing values include natural phenomena such as solar activity fluctuations, carbon emissions from natural sources like volcanoes, oceanic cycles, and more. Another key limitation is the issue of multicollinearity, where two or more independent variables are highly correlated with each other. This can inflate prediction errors and make it difficult to correctly ascertain the individual contribution of each multicollinear predictor. In this project, it is very likely that features such as CO_2 emissions and atmospheric CO_2 mean are highly correlated, just to name one example. By virtue of using a linear regression model, this analysis may also suffer from the fundamental assumption that linear relationships exist between the independent

variables and the dependent variable. Climate systems are inherently complex and volatile, containing drastic unpredictability and hypothesized inflection points (also known as 'tipping points'). Our simple linear model may fail to capture these nuances and may misrepresent the nature of the relationships between the features in the dataset.

In data science, ethical considerations are supremely important, as insights derived from data can be used to significantly influence decisions made by individuals, communities, and governments. Misrepresentation, biased modelling, or biased analysis can lead to poor decisions and erode trust in institutions and science as a whole. As regards this project, it brings together datasets from multiple sources, each with different original collection methods. Some of these original datasets may already suffer from sparse data collection, incorrect data (especially regarding data from centuries ago), and their own data preprocessing techniques. Then we must consider the effects of the data preprocessing applied to the merged dataset in this project. Using interpolation and other techniques to fill in values introduces synthetic data which may affect trend accuracy and skew model results, especially when applied to past. In climate science, the consensus is that the factors that drive it likely compound over time, so simple linear interpolation may have a massive adverse effect on model reliability.

Contributions of Artificial Intelligence to this Project

Artificial intelligence played a minor role in the project pipeline. The generative AI tool Gemini 2.5 supported this project primarily by providing suggestions on data visualization and how to best use the Matplotlib and Seaborn libraries. Its support was especially required to create the feature importance bar graph (Figure 2), as I had never used Seaborn before and was unaware of the F-Score metric for measuring feature importance.

Data collection, merging, and preprocessing, as well as model development, root cause analysis, and the entire creation of this report were done by me without any AI support.

References

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